Session Objectives

1. Assess the impact of upper respiratory infections (URIs) on the population and the effect on clinical practice
2. Identify the role of the immune system in combating URIs
3. Identify the common etiological agents of URIs
4. Identify the common etiological agents of community acquired pneumonia (CAP)
5. Compare and contrast nosocomial and CAP

Respiratory Tract Infections

- The vast majority of URIs are self-limited
- Viruses cause most cases of common cold, conjunctivitis, rhinitis, sinusitis, otitis, pharyngitis and acute bronchitis in non-smokers
- Viruses also cause many cases of pneumonia, but bacterial causes more common necessitating antibiotic therapy

Question

A 24-year-old woman presents to the clinic with a 3-day history of sore throat, nasal congestion, and mild fever. She denies any recent travel, known sick contacts, or significant medical history. On examination, she has erythema in the pharynx and nasal passages. There is no significant lymphadenopathy. A rapid antigen test for *Streptococcus pyogenes* is negative. Given her symptoms, she is likely experiencing a common upper respiratory infection (URI). Which of the following is the most important host defense mechanism responsible for preventing the invasion of pathogens in the respiratory tract?

A. Cough reflex
B. Gastric acidity
C. Alveolar macrophages
D. Surfactant
E. Normal flora

Question

A 5-year-old boy is brought to the clinic with a 2-day history of fever, sore throat, and difficulty swallowing. On examination, his throat is erythematous with mild swelling of the tonsils. His temperature is 101°F (38.3°C), and there is no significant lymphadenopathy. A throat culture grows *Streptococcus pneumoniae*. Which of the following is the most likely normal resident flora in the **pharynx** that may predispose this patient to this upper respiratory infection?

A. *Streptococcus pneumoniae*
B. *Staphylococcus epidermidis*
C. *Neisseria spp.*
D. *Corynebacterium*
E. *Peptostreptococcus*

Host Defenses

- Hairs within nares, ear
- Anatomy
- Cough reflex
- Swallow mechanism
- Gastric acidity
- Mucociliary clearance
- Alveolar macrophages
- Surfactant
- Normal flora

Disruption Due To

- CNS depression:
  - General anesthesia
  - Coma
- Neuromuscular disorders
  - Paralysis
  - ALS
  - Parkinson's
  - Anoxic brain damage
- Aspiration of gastric contents
  - Seizure
  - ETOH
- Foreign bodies
  - nasogastric tube
  - endotracheal intubation
- Antibiotics
- Cystic fibrosis
- Smoking
- Genetic ciliary defects
- Anoxia
- O_2 intoxication
- ARDS
- Trauma

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- Anoxia
- O_2 intoxication
- ARDS
- Trauma

Question

A 64-year-old male with a history of **smoking** and **chronic obstructive pulmonary disease (COPD)** is admitted to the hospital for a lower respiratory tract infection. On physical examination, his respiratory rate is 22 breaths per minute, and his oxygen saturation is 89%. He is intubated and placed on mechanical ventilation. The patient is also on antibiotics for the infection. Two days later, he develops worsening hypoxemia and chest X-ray reveals bilateral consolidation and interstitial infiltrates. Which of the following factors is most likely contributing to this patient's condition?

A. Normal ciliary clearance
B. Enhanced alveolar macrophage activity
C. Inhibition of mucociliary clearance due to smoking
D. Excessive gastric acidity
E. Use of antibiotics

Common Cold

- Most caused by rhinovirus
  - Also Coronaviridae, adenoviruses
  - Multiple antigenic types and limited duration of immunity allow colds throughout life
- Symptoms: coryza, rhinitis, sinus congestion, nasopharyngitis, cough, myalgias
- Extremely infectious
  - Droplets, person-to-person contact, contaminated surfaces
- No specific treatment (yet)

Question

A 27-year-old female presents to your clinic with complaints of a runny nose, sore throat, congestion, and mild body aches that started 2 days ago. She reports no fever and feels generally well, aside from a persistent cough and sinus pressure. On physical examination, she has mild nasal congestion, erythema of the throat, and clear nasal discharge. The rest of the examination is unremarkable. She works in an office environment and notes that several of her colleagues have had similar symptoms recently. Which of the following is the most likely cause of this patient's condition?

A. Streptococcal pharyngitis
B. Influenza
C. Common cold due to rhinovirus
D. Sinusitis
E. COVID-19

Sinusitis

- Etiologies:
  - Most commonly viral: rhinovirus, coronavirus
    - (Influenza, parainfluenza as part of their syndromes)
  - Bacterial: *S. pneumoniae*, *H. influenzae*
    - *Less commonly Moraxella catarrhalis, S. aureus, MRSA, Pseudomonas, Klebsiella*
  - Fungi unusual
    - Suspect immunocompromise

Sinusitis

- Clinical Presentation:
  - Sinus pain and congestion worsen with trunk flexion, nasal discharge
  - Sinus tenderness on examination
- Diagnosis
  - Clinical diagnosis based on presentation
  - Acute (<4 weeks) vs chronic (>12 weeks)
  - Viral versus bacterial versus fungal
  - Sinus computed tomography
  - Sinus aspiration or biopsy

Sinusitis: Treatment

- Acute: treat symptoms with antihistamines, nasal steroids
  - Suspect bacterial (e.g., high fever, prolonged course): amoxicillin or trimethoprim-sulfamethoxazole
- Chronic
  - Antihistamines, nasal steroids, saline irrigation
  - Treat bacterial infection if present with antibacterial therapy
  - Fix underlying defect (e.g., control blood sugar in patients with diabetes)
  - Surgery (e.g., osteotomy, polypectomy, debridement of rhinocerebral mucormycosis)

Question

A 35-year-old male presents to the clinic with a 2-week history of sinus pressure, nasal congestion, and thick, purulent nasal discharge. He also reports frontal headache, facial tenderness, and pain that worsens with bending forward. His symptoms started after a cold, and despite symptomatic treatment with over-the-counter decongestants, his condition has not improved. He denies fever or chills. On physical examination, there is tenderness to palpation over the maxillary sinuses and mild congestion of the nasal passages. What is the most likely diagnosis?

A. Acute viral sinusitis
B. Acute bacterial sinusitis
C. Chronic sinusitis
D. Allergic rhinitis
E. Nasal polyp

Acute Pharyngitis: Viral Etiology

Viruses	Notes
1. Rhinovirus (20% all cases) 2. Coronavirus (5% all cases)	Most common Associated with mild pharyngeal pain + URI symptoms
3. Influenza virus 4. Parainfluenza virus 5. Adenovirus	Variable severity Conjunctivitis in 33-50% of those with pharyngitis caused by adenovirus
6. Herpes simplex virus I & II 7. Cytomegalovirus 8. Epstein-Barr virus 9. HIV (acute infection) 10. Coxsackievirus A	Less common Consider in populations at risk

Acute Pharyngitis

- Streptococcus pyogenes
  - "FACE"
    - One point each for: Fever >= 100.4, anterior cervical Adenopathy, lack of Cough, tonsillar Exudate
      - **0-1**=symptomatic managemt, no culture
      - **2-3**=strep test and treat if positive
      - **4**=treat with penicillin (erythromycin if allergic) without testing

Question

A 13-year-old boy presents with a 2-day history of sore throat, fever (102°F), and difficulty swallowing. His mother reports that he has had a runny nose and mild cough for the past week. On examination, he has erythematous tonsils with white exudates, anterior cervical lymphadenopathy, and no cough. His vital signs are otherwise stable, and he is up-to-date on immunizations. Which of the following is the most likely cause of this patient's symptoms?

A. Group A Streptococcus (Streptococcus pyogenes)
B. Epstein-Barr virus (EBV)
C. Rhinovirus
D. Fusobacterium necrophorum
E. Coxsackievirus A

Laryngitis

- Often acute hoarseness, aphonia
- Laryngeal erythema + edema

Viruses (most common)	**Bacteria**	**Fungi**
Rhinovirus Coronavirus Influenza virus Parainfluenza virus Adenovirus Coxsackie virus RSV	*S. pyogenes* (Group A) *C. diphtheriae* (uncommon in US from widespread vaccination) *M. catarrhalis* *M. tuberculosis* (causes chronic laryngitis)	*Histoplasma* *Blastomyces* *Candida* *Coccidioides* *Cryptococcus*

- Treatment: Voice rest + humidifier use

Laryngeotracheobronchitis ("Croup")

- Etiology: 75% cases from Parainfluenza virus
  - adenovirus + RSV most of remainder
- Presentation:
  - 3 mo – 5 yo, "seal-barking" cough (↑ at night)
  - Variable severity of inspiratory stridor & respiratory distress
- Diagnosis: Clinical, viral culture / PCR
- Treatment: Supportive, 1.5-15% pts hospitalized

Question

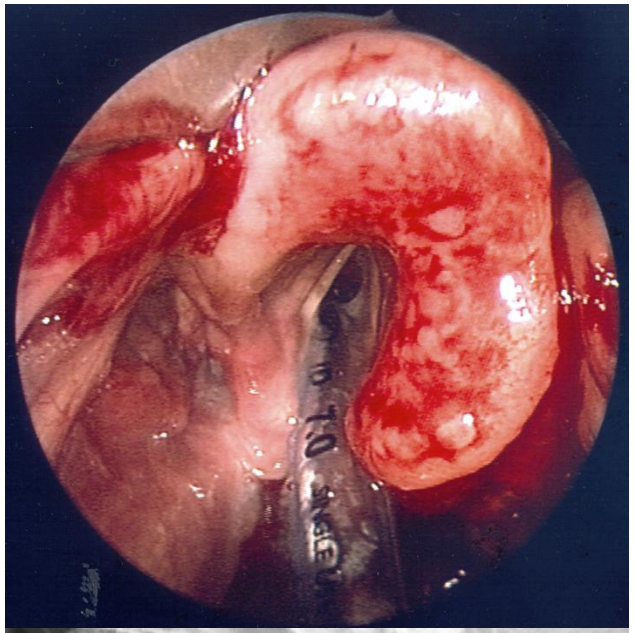
A 4-year-old boy presents to the emergency department with a **2-day history** of a **barking cough** that is worse at night. His mother notes that he has had a low-grade fever and mild difficulty breathing. On examination, he is agitated, with visible **inspiratory stridor** and a **hoarse voice**. His vital signs are as follows: temperature 101°F, heart rate 120 bpm, respiratory rate 28 breaths/min, and oxygen saturation 96% on room air. There is no significant nasal congestion or runny nose. The rest of his physical exam is unremarkable. Which of the following is the most likely diagnosis?

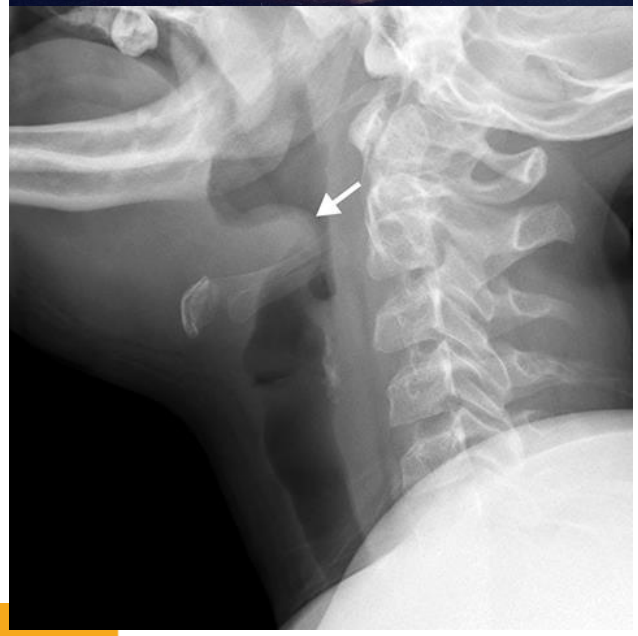
A. Acute laryngitis
B. Croup (Laryngeotracheobronchitis)
C. Bacterial tracheitis
D. Epiglottitis
E. Asthma exacerbation

Question

A 3-year-old boy presents to the emergency department with a **4-day history of a cough** that has become progressively worse, particularly at night. His mother reports that the cough sounds like a "seal bark," and the child has had difficulty breathing, with noticeable **inspiratory stridor**. On examination, he is **febrile** and appears distressed, but is not in significant respiratory failure. His oxygen saturation is normal, and there is no history of recent respiratory infections. What is the most likely diagnosis for this child's condition?

A. Acute laryngitis
B. Bacterial tracheitis
C. Laryngeotracheobronchitis (Croup)
D. Foreign body aspiration
E. Bronchiolitis





Epiglottitis

Etiology:

- Often H. influenzae, type b
- A Gram -ve short bacillus
- ↓ incidence with widespread Hib vaccination, except possibly in very underdeveloped parts of the world

Dx:

- Peds > adults
- Fever, severe sore throat, head-forward carriage, drooling, stridor, dyspnea,
- X-ray: thumb print sign

Tx:

- Secure airway immediately!
- IV ampicillin/sulbactam, cefuroxime, or ceftriaxone; TMP-SMX if PCN allergic

Prevention:

- Hib vaccine

Question

A 3-year-old boy is brought to the emergency department by his parents with a **2-day history** of worsening **sore throat** and **fever**. The child has been **drooling** and has been sitting **leaning forward** in a **tripod position** since the symptoms started. His parents report that he is **struggling to breathe** and is having difficulty swallowing. On physical exam, the child is febrile (102°F), agitated, and has **inspiratory stridor**. **Drooling** is noted, and he is in a **forward-leaning** posture. **X-ray** shows the characteristic **thumbprint sign**. Which of the following is the most likely diagnosis?

A. Acute viral pharyngitis
B. Croup (laryngotracheobronchitis)
C. Epiglottitis
D. Bacterial tracheitis
E. Foreign body aspiration

Question

A 55-year-old male with a history of **hypertension** and **smoking** presents to the emergency department with a **2-day history** of **fever**, **cough**, **dyspnea**, and **pleuritic chest pain**. He reports feeling **chilled** and **fatigued** but denies any recent travel or sick contacts. On physical examination, his **temperature** is 101.2°F, **heart rate** is 105 bpm, **respiratory rate** is 22 breaths per minute, and his **oxygen saturation** is 92% on room air. He has **crackles** on lung auscultation, and there is a **pleural friction rub** present. A **chest X-ray** shows **acute consolidation** in the right lower lobe. Which of the following is the most likely diagnosis?

A. Acute bronchitis
B. Pulmonary embolism
C. Community-acquired pneumonia
D. Acute exacerbation of chronic obstructive pulmonary disease (COPD)
E. Lung abscess

CAP: Patient Approach

History of Present Illness	When did it begin?, any factors exacerbate or alleviate chief complaint?, quality?, radiation (if pain)?, severity? (also ask about effects on activities of daily living to gauge severity), timing
Past Medical History	COPD, asthma, immunosuppressive diseases or states, aspiration risk factors, etc.
Medications/allergies	Chronic use of immunosuppressants (e.g. steroid), bronchodilators, CNS depressants (barbiturates, benzodiazepines)
Social History	Cigarette smoking, alcoholism, IV drug abuse, aspiration risk factors, occupation, living arrangements, recent travel, exposure to sick individuals, exposure to animals, vaccination history, recent antibiotic use (within last 30 days)
Review of Systems	Constitutional - fever/chills/night sweats Respiratory - cough, dyspnea, tachypnea MSK - arthralgia, myalgia

Question

A 60-year-old male with a history of **chronic obstructive pulmonary disease (COPD)** and **hypertension** presents to the emergency department with a **2-day history** of **fever**, **productive cough**, and **shortness of breath**. He reports that his symptoms started **gradually** and have worsened over the past 24 hours. The cough is **associated with yellow-green sputum**, and he notes **mild pleuritic chest pain** when coughing. He denies any recent travel, sick contacts, or exposure to animals. He admits to a 40-pack-year smoking history and **occasionally uses inhalers** for his COPD. On physical examination, his temperature is 101°F, his heart rate is 110 bpm, and his respiratory rate is 22 breaths per minute. He has **crackles** on auscultation of the right lower lung field, and his oxygen saturation is 92% on room air. A **chest X-ray** shows **right lower lobe consolidation**. Which of the following is the most appropriate next step in the management of this patient?

A. Initiate oral antibiotics based on outpatient guidelines
B. Begin intravenous (IV) antibiotics and admit for observation
C. Perform bronchoscopy to confirm the diagnosis
D. Order a CT scan of the chest to assess for lung masses
E. Recommend outpatient observation and symptomatic treatment

CAP epidemiologic factors

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Factor	Possible Pathogen(s)
Alcoholism	*S. pneumoniae,* oral anaerobes, *K. pneumoniae, Acinetobacter* spp., *M. tuberculosis*
COPD and/or smoking	*H. influenzae, P. aeruginosa, Legionella* spp., *S. pneumoniae, M. catarrhalis, Chlamydia pneumoniae*
Structural lung disease (e.g., bronchiectasis)	*P. aeruginosa, Burkholderia cepacia, S. aureus*
Dementia, stroke, decreased level of consciousness	Oral anaerobes, Gram -ve enterics
Lung abscess	CA-MRSA, oral anaerobes, endemic fungi, *M. tuberculosis*, atypical *Mycobacteria*

Question

A 68-year-old male with a history of **chronic obstructive pulmonary disease (COPD)** and a 40-pack-year smoking history presents to the emergency department with a **2-day history** of **fever**, **cough**, **dyspnea**, and **purulent sputum**. He also reports **pleuritic chest pain** and mild **shortness of breath** that has worsened over the last 24 hours. He denies any recent travel, sick contacts, or exposure to animals. On physical examination, his temperature is 101°F, heart rate is 108 bpm, and respiratory rate is 22 breaths per minute. Crackles are heard in the right lower lung field, and his oxygen saturation is 92% on room air. A **chest X-ray** reveals **right lower lobe consolidation**. Given this patient's history and presentation, which of the following pathogens is most likely responsible for his community-acquired pneumonia (CAP)?

A. **Klebsiella pneumoniae**
B. **Pseudomonas aeruginosa**
C. **Legionella spp.**
D. **Mycobacterium tuberculosis**
E. **Streptococcus pneumoniae**

Question

A 32-year-old male presents to the emergency department with a **fever**, **cough**, **dyspnea**, and **pleuritic chest pain**. He reports a recent trip to the **Ohio River Valley** for a weekend camping trip, where he had close exposure to **bats** in an abandoned building. His symptoms began a few days after returning home. On examination, his temperature is 101°F, respiratory rate is 22 breaths per minute, and he appears mildly tachypneic. Chest X-ray reveals a **patchy infiltrate** in the **right lower lobe**. Which of the following pathogens is most likely responsible for this patient's pneumonia?

A. **Histoplasma capsulatum**
B. **Hantavirus**
C. **Legionella pneumophila**
D. **Chlamydia psittaci**
E. **Coxiella burnetii**

CAP: Microbiologic Diagnosis

- Sputum culture & Gram staining rarely done
- Blood culture (in all hospitalized patients)
- Ag tests
  - *Legionella, S. pneumoniae,* influenza, RSV, SARS-CoV-2
- PCR
  - Influenza, SARS-CoV-2, *M. tuberculosis*
- Serology
  - Cold agglutination - *Mycoplasma*

Question

A 65-year-old male with a history of **chronic obstructive pulmonary disease (COPD)** presents to the emergency department with a **fever**, **productive cough**, **dyspnea**, and **pleuritic chest pain** for the past 3 days. He has **tachypnea** and **tachycardia** on examination, with **crackles** heard on lung auscultation. A **chest X-ray** shows **lobar consolidation** in the right lower lobe. Given his **age** and **underlying COPD**, you decide to admit him to the hospital. Which of the following microbiologic tests would be most useful for diagnosing the pathogen responsible for his pneumonia?

A. **Sputum Gram stain and culture**
B. **Blood cultures**
C. **Legionella urinary antigen test**
D. **Cold agglutinin test**
E. **SARS-CoV-2 PCR**

CAP: Treatment

- Outpatient therapy
  - No comorbidies, MRSA or *Pseudomonas* risk: amoxicillin or doxycycline or macrolide
  - Cormobidities
    - combination of amoxicillin/clavulanate or cephalosporin *and* macrolide

    OR
    - Quinolone
  - Consider antibiotic resistance (especially macrolides)
- Inpatient beta-lactam + macrolide OR fluoroquinolone
  - Add MRSA and/or *Pseudomonas* coverage if at risk

Question

A 72-year-old woman with a history of **hypertension**, **type 2 diabetes**, and **chronic obstructive pulmonary disease (COPD)** presents to the emergency department with a **3-day history** of **fever**, **productive cough**, and **dyspnea**. On examination, she has **tachypnea**, **tachycardia**, and **crackles** in the right lower lung field. A **chest X-ray** shows **right lower lobe consolidation**, consistent with **community-acquired pneumonia (CAP)**. Blood cultures are sent. Given her **age**, **COPD**, and **comorbidities**, which of the following is the most appropriate outpatient treatment for her pneumonia?

A. **Amoxicillin**
B. **Amoxicillin/clavulanate and macrolide**
C. **Levofloxacin**
D. **Doxycycline**
E. **Ceftriaxone and azithromycin**

Question

A 65-year-old man with a history of **chronic obstructive pulmonary disease (COPD)** and **type 2 diabetes mellitus (DM)** presents to your clinic for a routine follow-up. He is a **smoker** with a 40 pack-year history and is concerned about his risk of respiratory infections. He has not received any recent vaccinations and is unsure about which vaccines he should receive to help prevent **community-acquired pneumonia (CAP)**.

Which of the following vaccinations is most appropriate for this patient to reduce his risk of pneumonia?

A. **Influenza vaccine**
B. **Pneumococcal polysaccharide vaccine (PPV23)**
C. **Pneumococcal conjugate vaccine (PPV13)**
D. **Both PPV23 and PPV13**
E. **Varicella vaccine**

Question

Question:
A 22-year-old otherwise healthy woman presents to the emergency department with a 5-day history of **dry cough**, **mild myalgia**, and **headache**. She reports feeling mildly fatigued but denies chest pain, difficulty breathing, or productive sputum. On physical examination, she is afebrile, and her lungs auscultate clear with no signs of consolidation. A chest X-ray reveals a **patchy infiltrate** that appears more extensive than her clinical presentation suggests.

Which of the following is the most likely cause of her symptoms?

A. **Streptococcus pneumoniae**
B. **Legionella pneumophila**
C. **Klebsiella pneumoniae**
D. **Pneumocystis jirovecii**
E. **Mycoplasma pneumoniae**

Summary

- Impact of upper respiratory infections (URIs) on the population and the effect on clinical practice
- Role of the immune system in combating RTIs
- Common etiological agents of RTIs
- Common etiological agents of community acquired pneumonia (CAP)
- Compared and contrasted nosocomial and CAP
- Compared and contrasted typical and atypical CAP